

Texture-invariant conformal prediction for compound-Gaussian autoregressive episodes:

exact laws, noise floor, and history-binned calibration
R7 working theory (branch `research/r7-clutter`)

Draft for author review

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Status. Working mathematical draft. Every numbered statement has a numerical check in `research/r7c/validation`. Known ingredients are cited, not claimed: Gaussian quadratic-form distributions; the CA-CFAR threshold law; conformal rank validity; pivot-based conditional validity. Proofs require independent human review before manuscript use.

1 Model and procedures

An episode is $X = (H_1, \dots, H_m, Y) \in \mathbb{C}^{m+1}$. Conditional on a latent texture $\tau > 0$,

$$X \mid \tau \sim \mathcal{CN}(0, \nu(\tau R + I)), \quad \nu \geq 0, \quad (1)$$

where R is the unit-diagonal clutter correlation matrix and ν the thermal-noise power. $\nu = 0$ with the convention $\nu\tau \equiv \sigma^2$ gives the pure compound-Gaussian (SIRV) model $X \mid \sigma \sim \mathcal{CN}(0, \sigma^2 R)$. We write $c = \tau$ for the clutter-to-noise ratio (CNR). For AR(1) clutter, $R_{jk} = \rho^{j-k}$ for $j \geq k$ and $R_{kj} = \overline{R_{jk}}$.

A *linear normalized score* is specified by a coefficient vector $a \in \mathbb{C}^m$ (the centre $a^\top H$) and a positive semidefinite $m \times m$ matrix Q (the history scale $s(H)^2 = H^\mathsf{H}QH/m$):

$$S(X) = \frac{|Y - a^\top H|}{s(H)}. \quad (2)$$

IN-ARCP (R6, complexified) uses $a = re_m$ and $Q = W_r^\mathsf{H}W_r$, where W_r is the bidiagonal whitening matrix with first diagonal entry $\sqrt{1 - |r|^2}$. AR(p) conditional-innovation scores use the AR(p) prediction for a and $Q = V^\mathsf{H}V/(m-p) \cdot m$, where V maps H to its $m-p$ conditional innovations. Split conformal calibration uses the k th smallest of n calibration scores, $k = \lceil (n+1)(1-\alpha) \rceil$.

2 Exact score law

Proposition 1 (Elementary exact coverage). *Let $X \sim \mathcal{CN}(0, \Sigma)$ with $\Sigma \succ 0$. Fix (a, Q) and $q > 0$. Let $b = e_{m+1} - (a^\top, 0)^\top$, where e_{m+1} selects Y , and*

$$M_q = \bar{b}b^\top - \frac{q^2}{m} \begin{pmatrix} Q & 0 \\ 0 & 0 \end{pmatrix}, \quad \lambda_1, \dots, \lambda_{m+1} = \text{eig}(\Sigma^{1/2}M_q\Sigma^{1/2}).$$

If the nonzero λ_j are distinct, then

$$G_\Sigma(q) := \mathbb{P}\{S(X) \leq q\} = 1 - \sum_{j: \lambda_j > 0} \prod_{k \neq j, \lambda_k \neq 0} \frac{\lambda_j}{\lambda_j - \lambda_k}. \quad (3)$$

Proof. $\{S \leq q\} = \{X^H M_q X \leq 0\}$. Write $X = \Sigma^{1/2} \xi$ with $\xi \sim \mathcal{CN}(0, I)$ and diagonalize: $X^H M_q X = \sum_j \lambda_j |\eta_j|^2$ with $|\eta_j|^2$ iid $\text{Exp}(1)$. The characteristic function is $\prod_j (1 - it\lambda_j)^{-1}$. Partial fractions in t give density $\sum_j |\lambda_j|^{-1} e^{-x/\lambda_j} \mathbf{1}\{x/\lambda_j > 0\} \prod_{k \neq j} \lambda_j / (\lambda_j - \lambda_k)$. Integrating over $x > 0$ gives (3). Zero eigenvalues contribute factors $\lambda_j / (\lambda_j - 0) = 1$. $\square$

Remark 1. M_q has exactly one positive eigenvalue when $Q \succ 0$, because $\bar{b}b^T$ has rank one and the other block is negative definite on the history coordinates. So (3) has a single term. The real-valued R6 law needs a sphere integral instead. At the true AR(1) coefficient with pure clutter, (3) reduces to the $F(2, 2m)$ law $1 - (1 + q^2/m)^{-m}$. That is the Gaussian CA-CFAR threshold law, a known result. Repeated eigenvalues are handled by continuity or by Gil-Pelaez inversion. Numerical check: 144 cases against 10^6 -draw Monte Carlo and against Gil-Pelaez (agreement 4×10^{-10}), in `v1_exact_laws.py` and `v1b_check_worst.py`.

Corollary 1 (Texture-conditional exactness without noise; drift with noise). *Under (1) with $\nu = 0$, $G_{\sigma^2 R}(q) = G_R(q)$ for every $\sigma > 0$. So any split-conformal threshold computed from exchangeable calibration episodes gives coverage, conditional on the test texture, equal to $\mathbb{E} G_R(\hat{q})$, which is at least $1 - \alpha$. With $\nu > 0$, the conditional coverage at CNR c is the explicit function*

$$\mathcal{C}(c; \hat{q}) = G_{cR+I}(\hat{q}), \quad \lim_{c \rightarrow \infty} \mathcal{C}(c; q) = G_R(q), \quad \lim_{c \rightarrow 0} \mathcal{C}(c; q) = G_I(q).$$

Proof. The eigenvalues in (3) scale by σ^2 , which leaves the product ratios and signs unchanged. With noise, ν cancels in the same way, leaving $cR + I$. The limits follow from continuity of eigenvalues after dividing by c (as $c \rightarrow \infty$), and directly (as $c \rightarrow 0$). $\square$

The drift range $|G_R(\hat{q}) - G_I(\hat{q})|$ is therefore computable before deployment from $(a, Q, R, \hat{q})$. Check: `v1_exact_laws.py` (V1b) gives the curve $0.883 \rightarrow 0.915$ over -10 to 30 dB.

3 Expected coverage under calibration–test shift

Proposition 2. *Let the calibration scores be iid with continuous CDF G_{cal} , and the test score independent with CDF G_{test} (same (a, Q)). Then*

$$\mathbb{E}[G_{\text{test}}(\hat{q})] = \int_0^\infty G_{\text{test}}(q) dI_{G_{\text{cal}}(q)}(k, n+1-k),$$

where I_z is the regularized incomplete beta function. If both laws come from (1), possibly with different (R, ν) or texture mixtures, the integrand is explicit through Proposition 1 averaged over the texture distribution.

Proof. $\hat{q} = G_{\text{cal}}^{-1}(B)$ with $B \sim \text{Beta}(k, n+1-k)$ (R6, eq. (8)). So $\mathbb{P}(\hat{q} \leq q) = I_{G_{\text{cal}}(q)}(k, n+1-k)$. Condition on $\hat{q}$ and integrate. $\square$

Check: the exact value matches 4,000-replicate Monte Carlo within one standard error in three shift cases (V1c). On 182 real IPIX session pairs, predictions correlate 0.75 with observed coverage (V3). It is an explanatory tool, not a certificate.

4 Noise-aware normalization and its zero-noise limit

Given (R, ν) , estimate the CNR from the history alone:

$$\hat{c}(H) = \arg \max_{c \geq 0} \left\{ -\log \det(cR_H + I) - \nu^{-1} H^H (cR_H + I)^{-1} H \right\}.$$

Then use the Gaussian predictive centre $\mu(H) = \Sigma_{YH} \Sigma_{HH}^{-1} H$ and variance $v(H) = \nu \{\hat{c} + 1 - \hat{c}^2 r_{HY}^H (\hat{c}R_H + I)^{-1} r_{HY}\}$ at $c = \hat{c}(H)$. The NA score is $|Y - \mu(H)| / \sqrt{v(H)}$, calibrated conformally.

Proposition 3 (Reduction to IN-ARCP). *Let R be $AR(1)$ with coefficient ρ , and fix a history $H \neq 0$. As $\nu \downarrow 0$: $\nu\hat{c}(H) \rightarrow \hat{\sigma}^2(H) := H^H R_H^{-1} H / m$, $\mu(H) \rightarrow \rho H_m$ and $v(H) \rightarrow s_\rho(H)^2$. Hence the NA score converges pointwise to the IN-ARCP score at $r = \rho$.*

Proof sketch. Put $\gamma = \nu c$. The objective becomes $-\log \det(\gamma R_H + \nu I) - H^H(\gamma R_H + \nu I)^{-1} H$ plus a constant. It converges locally uniformly on $\gamma \in (0, \infty)$ to $-m \log \gamma - \gamma^{-1} H^H R_H^{-1} H$ plus a constant, whose unique maximizer is $\hat{\sigma}^2(H)$. Coercivity at 0 and ∞ gives convergence of the maximizers. For $AR(1)$, $\Sigma_{YH} \Sigma_{HH}^{-1} \rightarrow \rho e_m^T$ (Markov property). The predictive variance tends to $\gamma(1 - |\rho|^2)$, and $(1 - |\rho|^2)R_H^{-1} = W_\rho^H W_\rho$ gives $\hat{\sigma}^2(1 - |\rho|^2) = s_\rho(H)^2$. $\square$

On real IPIX data, NA gives 12–33% smaller discs at equal coverage in noise-affected sessions and is neutral in clean ones (V2b). Its Gaussian plug-in under-covers (0.874 at $\alpha = 0.1$), which conformal calibration corrects.

5 History-binned (Mondrian) calibration under a latent texture

Let $T(H) > 0$ be homogeneous of degree one, $T(\sigma h) = \sigma T(h)$; the history scale $s(H)$ is an example. Mondrian calibration assigns bins $B_b = [e_{b-1}, e_b)$, $0 = e_0 < \dots < e_K = \infty$, and uses the bin-wise $(1 - \alpha)$ calibration quantile q_b . Its validity is $\mathbb{P}(S \leq q_b \mid T \in B_b) \geq 1 - \alpha$, which is conditional on the *observed* bin.

Proposition 4 (Mondrian tilt). *Assume pure SIRV ($\nu = 0$), with $X = \sigma Z$, $\sigma \sim G$ independent of Z , and population bin quantiles q_b . Let $\mathcal{C}(\sigma) = \sum_b \mathbb{P}\{S(Z) \leq q_b, \sigma T(Z_H) \in B_b\}$ be the coverage conditional on texture.*

1. *If $S(Z)$ is independent of $T(Z_H)$, then $\mathcal{C}(\sigma) = 1 - \alpha$ for all σ .*
2. *If the conditional law of $S(Z)$ given $T(Z_H) = t$ is strictly stochastically decreasing in t , and G is non-degenerate, then*

$$\lim_{\sigma \rightarrow \infty} \mathcal{C}(\sigma) < 1 - \alpha < \lim_{\sigma \rightarrow 0} \mathcal{C}(\sigma).$$

The unbinned conformal threshold instead gives $\mathcal{C}(\sigma) \equiv \text{constant}$.

The condition in (2) holds exactly for IN-ARCP at the true coefficient with $T = s_\rho$: then $S = |E|/s_\rho(Z_H)$ with the innovation E independent of Z_H .

Proof. (1) Independence makes every bin-conditional law of S equal to its marginal law. Hence $q_b \equiv q$ and $\mathcal{C}(\sigma) = \mathbb{P}(S \leq q)$.

(2) As $\sigma \rightarrow \infty$, $\mathbb{P}\{\sigma T(Z_H) \in B_K\} \rightarrow 1$, so $\mathcal{C}(\sigma) \rightarrow F_S(q_K)$. Now q_K is the $(1 - \alpha)$ quantile of $S(Z)$ given $\sigma T(Z_H) \geq e_{K-1}$ under the mixture over σ . That event selects larger values of $T(Z_H)$ than the unconditional law, strictly so when G is non-degenerate and $e_{K-1} > 0$. By the monotone likelihood-ratio ordering of $T(Z_H)$ given the event, and stochastic monotonicity of S given T , the conditional law of S in the top bin is strictly stochastically smaller than its marginal law. So $q_K < F_S^{-1}(1 - \alpha)$ and $F_S(q_K) < 1 - \alpha$. The case $\sigma \rightarrow 0$ is symmetric with the bottom bin. For IN-ARCP at $r = \rho$, $S = |E|/T$ with $E \perp T$, so $\mathbb{P}(S \leq x \mid T = t) = \mathbb{P}(|E| \leq xt)$ is strictly increasing in t . $\square$

Remark 2. This contradicts the routine recommendation to bin on estimated difficulty (Boström and Johansson 2020; Dewolf et al. 2025) *when the target is coverage conditional on a latent nuisance*, which is the CFAR-type requirement in radar. Mondrian remains valid for its own target, the observed bin. Empirical Bayes shrinkage of the texture shows the same trade-off numerically (V2a). Checks: synthetic spread $0.037 \rightarrow 0.074$ (Mondrian); IPIX tilt $0.924 \rightarrow 0.888$ (V2b).

6 Open items

- Monotonicity in c of $\mathcal{C}(c; q)$ holds numerically; no proof yet.
- A local expansion of expected size for the complex AR(p) case is not derived. The exact integral of Proposition 2-type suffices for the paper.
- A rigorous version of the limit step in Proposition 4(2) needs the MLR ordering argument written out for general G ; checked numerically only for lognormal G .